N+ NETWORK FUNDAMENTALS BUILDING NETWORK

UNDERSTANDING AND MAINTANCE OF NETWORK

DAY4

TASKS:1

List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

RESULT

Ethernet Cable	**Fiber Optic Cable**
Uses electrical signals to transmit data.	Uses light signals to transmit data.
Lower speed and shorter distance (up to 100 meters).	Very high speed and supports long distances (several kilometers).
Less expensive and easier to install.	More expensive and requires special tools for installation.

EXAMPLE:-

- **Ethernet Cable:** Used to connect a home Wi-Fi router to a desktop computer.
- **Fiber Optic Cable:** Used as the network backbone in colleges, data centers, and Internet Service Providers (ISPs)

TASKS:2

Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.

RESULT

Pin	Wire Color
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

TASKS:3

Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

RESULT

1. Cut the CAT6 cable to the required length.
2. Use a cable stripper to remove about 2–3 cm of the outer jacket.

3. Untwist the wire pairs carefully.
4. Arrange the wires according to the **T568B** color order.
5. Straighten and trim the wires evenly.
6. Insert the wires into the RJ45 connector until each wire reaches the end.
7. Place the connector into the RJ45 crimping tool.
8. Press the crimper firmly to secure the connector.
9. Test the cable using a network cable tester.

TASKS:4

Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences. Consider speed, distance, and budget for your answer.

RESULT

For gaming café I would use **CAT6 Ethernet cables** because all the computers are in the same room, making Ethernet fast, reliable, and affordable. I would use an **RJ45 crimper** to attach RJ45 connectors to the CAT6 cables. Fiber optic cables are not necessary because they are more expensive and mainly used for long-distance communication. Ethernet provides low latency, which is ideal for multiplayer gaming.